Exercise session n°3 : stochastic bandits (part 2)

Exercise 1 :

Concentration for sequences of random length. Let $X_1, X_2, \ldots$ be a sequence of independent standard Gaussian random variables defined on probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $T : \Omega \to \{1, 2, 3, \ldots\}$ is another variable and let $\hat{\mu}_T = \sum_{t=1}^{T} \frac{X_t}{T}$ be the empirical mean based on T samples.

1) Show that if T is independent from X_t for all t, then for any $\delta \in (0, 1)$

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) \leq \delta.$$

2) Now relax the assumption that T is independent from $(X_t)_t$. Let $E_t = \mathbb{1}_{T=t}$ and $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ be the σ-algebra generated by the first t samples. Let $\delta \in (0, 1)$ and show there exists a random variable T such that for all t, E_t is $\mathcal{F}_t$-measurable and

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) = 1.$$

Hint: You can use the law of the iterated logarithm, which says if $X_1, X_2, \ldots$ is a sequence of independent and identically distributed random variables with zero mean and unit variance, then

$$\limsup_{n\to\infty} \frac{\sum_{t=1}^{n} X_t}{\sqrt{2n \ln\ln n}} = 1 \quad \text{almost surely.}$$

3) What is the relation between the above inequality and our concentration lemma for the empirical means in bandits problems? Do **2)** and our lemma contradict? Why?

Exercise 2 :

Phased SE. Consider the following phased Successive Eliminations algorithm parameterized by $a > 1$.

1) Show a regret bound similar to Successive Eliminations algorithm.

2) What is the role played by a?

Exercise 3 :

Adapting to reward variance. Let $X_1, \ldots, X_N$ be a sequence of i.i.d. random variables

Algorithm: Phased Successive Eliminations
input: T, $a \geq 1$
$\mathcal{K} \leftarrow [K]$
$\ell \leftarrow 0$
while $\mathrm{Card}(\mathcal{K}) > 1$ **do**
  pull all arms in $\mathcal{A}$ $\lceil a^\ell \rceil$ times
  for *all* $k \in \mathcal{K}$ *such that* $\hat{\mu}_k + \sqrt{\frac{2\ln T}{N_k(T)}} \leq \max_{i\in\mathcal{K}} \hat{\mu}_i - \sqrt{\frac{2\ln T}{N_i(T)}}$ **do** $\mathcal{A} \leftarrow \mathcal{A} \setminus \{k\}$
  $\ell \leftarrow \ell + 1$
repeat pull only arm in $\mathcal{K}$ **until** $t = T$

with mean μ, variance σ^2 and bounded support so that $X_t \in [0, M]$ almost surely. Define the estimators

$$\hat{\mu}_N = \frac{1}{N}\sum_{t=1}^{N} X_t$$

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N} (\hat{\mu} - X_t)^2.$$

We admit in the following the **empirical Bernstein** inequality:

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\hat{\sigma}^2}{N}\ln(3/\delta)} + \frac{3M}{N}\ln(3/\delta)\right) \leq \delta.$$

1) Show that the Bernstein inequality in the course implies here

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\sigma^2}{N}\ln(2/\delta)} + \frac{2M}{3N}\ln(2/\delta)\right) \leq \delta$$

Comment on the differences between the two above *Bernstein inequalities*.

2) Show that $\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N}(X_t - \mu)^2 - (\hat{\mu}_N - \mu)^2$.

3) Is $\hat{\sigma}^2$ an unbiased estimator of σ^2? If not, can we easily make it unbounded?

4) Show that

$$\mathbb{P}\left(\hat{\sigma}^2 \geq \sigma^2 + \sqrt{\frac{2M^2\sigma^2}{N}\ln(1/\delta)} + \frac{2M^2}{3N}\ln(1/\delta)\right) \leq \delta.$$

Hint: Use Bernstein inequality of **1)**.

5) **(Hard)** Consider a bandit setting with K arms, bounded rewards $X_k(t) \in [0, M]$ and the variance of the k-th arm is σ_k^2. Design a policy that depends on M, but does not need to know

σ_i a priori, such that there exists a universal constant $C > 0$ with

$$R_T \leq C \sum_{k,\Delta_k>0} \left(\Delta_k + (M + \frac{\sigma_i^2}{\Delta_i}) \ln T \right).$$

Hint: Without a complete proof, the empirical Bernstein inequality can be extended (up to some changes) to cases where N is a random variable.

Exercise 4 :

This exercise studies the celebrated Thompson sampling algorithm and aims at proving an optimal instance dependent regret bound on the problem of Bernoulli rewards. The Thompson sampling algorithm is described below.

In words, Thompson sampling starts with a prior distribution $\boldsymbol{p}_0$ distribution on the (mean) parameters of the bandits instance and at each round t, it draws random samples $\theta_k(t)$ from the posterior distribution $\boldsymbol{p}_{t-1}$ on the instance parameters at time $t-1$, which is defined as

$$\boldsymbol{p}_{t-1}(A) = \mathbb{P}\Big((\mu_1, \dots, \mu_K) \in A \mid \mathcal{F}_{t-1} \Big) \quad \text{for any } A \in B(\mathbb{R}), \tag{1}$$

where $\mathcal{F}_{t-1} = \sigma\Big(U_1, X_{a_1}(1), U_2, X_{a_2}(2), \dots X_{a_{t-1}}(t-1) \Big)$ and the U_s are random variables uniformly drawn in $[0,1]$, that are independent with all other variables.

Algorithm: Thompson sampling
input: prior distribution $\boldsymbol{p}_0$
for $t = 1, \dots, T$ **do**
  Sample $\boldsymbol{\theta}(t) \sim \boldsymbol{p}_{t-1}$
  Pull $a_t \in \operatorname{argmax}_{k\in[K]} \theta_k(t)$ `// Ties broken arbitrarily`
  Update $\boldsymbol{p}_t$ as the posterior distribution of the parameters, following Bayes rule.

We note for each time $t \in \mathbb{N}$ and arm $k \in [K]$:

$$S_k(t) = \sum_{s=1}^{t} X_k(s) \mathbb{1}_{a_s = k}.$$

1) Consider an instance of Bernoulli bandits, i.e., $\mathcal{D} = \{\text{Bernoulli}(\mu) \mid \mu \in [0,1]\}^K$. Show then that in the case of Bernoulli rewards with a uniform prior, at each time $t \in \mathbb{N}$, $\boldsymbol{p}_{t-1}$ is the joint distribut1ion of K independent Beta distributions, where the k-th Beta distribution has parameters $(S_k(t-1)+1, N_k(t-1) - S_k(t-1) + 1)$. In other words for any $t \in \mathbb{N}$, the drawn samples $\theta_k(t)$ are independent with each other conditioned on $\mathcal{F}_{t-1}$ and

$$\theta_k(t) \sim \text{Beta}(S_k(t-1)+1, N_k(t-1) - S_k(t-1) + 1).$$

2) Consider now that the prior is the improper uniform distribution[1] on $\mathbb{R}$ and Gaussian bandits with variance σ^2, i.e., $\mathcal{D} = \{\mathcal{N}(\mu, \sigma^2) \mid \mu \in \mathbb{R}\}^K$.
For any $t \in \mathbb{N}$, what is the distribution of $\boldsymbol{p}_{t-1}$ in this case?

[1] This can be seen as the uniform distribution on $\mathbb{R}$. It is not a proper distribution, since it is not of measure 1, but the Bayes rule can still be applied with it.